

ABHIUDAYA PRATAP SINGH

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Professional Summary

Computer Science Engineering student specializing in AI and Machine Learning, with hands-on experience designing, training, and deploying predictive models using Python and Scikit-learn. Skilled in data preprocessing, feature engineering, and model evaluation, with additional experience in full-stack web development using React.js and Node.js. Exposure to Generative AI concepts including Prompt Engineering, RAG, and LLM fundamentals.

Education

B.Tech in Computer Science Engineering	2023 – 2027
Vellore Institute of Technology, Bhopal	CGPA: 8.59/10
Senior Secondary (12th)	2023
Army Public School, Lucknow	Percentage: 77.5%
Secondary School (10th)	2021
Army Public School, Lucknow	Percentage: 90.0%

Technical Skills

- Programming Languages:** Python, Java, JavaScript, SQL
- Machine Learning & AI:** Machine Learning, Data Analysis, Predictive Modeling, Scikit-learn, Pandas, NumPy
- Generative AI:** Prompt Engineering, Retrieval-Augmented Generation (RAG), Large Language Model (LLM) Fundamentals
- Web Development:** HTML, CSS, React.js, Node.js, REST APIs
- Tools & Databases:** Git, GitHub, VS Code, MySQL

Experience

Machine Learning and Full Stack Developer (Project-Based)	Jan 2024 – Present
Madhya Pradesh, India	

- Designed and implemented supervised machine learning models (Logistic Regression, Random Forest) for healthcare risk prediction systems, achieving up to [80]% prediction accuracy across a dataset of [1000] patient records.
- Performed data cleaning, preprocessing, and feature engineering on [1,000]+ data points, improving model accuracy by [15]% and reducing prediction error rate.
- Built and deployed a full-stack web application integrating ML models with a React.js frontend and backend APIs, reducing prediction response time to under [2] seconds.
- Enabled real-time prediction functionality through efficient backend-to-frontend communication, supporting [30]+ concurrent user requests during testing.

Projects

Diabetes Prediction APP	Jan 2024
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Tech Stack: Python, Scikit-learn, Pandas, NumPy, HTML, CSS, Streamlit
GitHub — Live Demo

- Developed a machine learning-based system to assess diabetes risk using key medical parameters.
- Implemented Logistic Regression classification and optimized performance using precision and F1-score metrics.
- Applied data preprocessing, feature selection, and model validation to improve prediction reliability.
- Built and deployed an interactive web interface for instant health risk predictions.

Smart Health AI	Jul 2026 – Jul 2026
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Tech Stack: React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, Google Gemini API
GitHub — Live Demo

- Designed and developed a full-stack healthcare platform enabling users to access AI-powered health assistance through a secure, interactive web application.
- Integrated the Google Gemini API to deliver intelligent, health-related responses and real-time assistance to users.
- Implemented a JWT-based authentication system for secure user registration and login.
- Built a scalable REST API backend with Node.js and Express.js, paired with a responsive React.js frontend for a seamless user experience.
- Utilized MongoDB with Mongoose for efficient storage and management of user data.
- Developed a clean, modern UI/UX for intuitive, seamless interaction with the platform.

Certifications

- Deloitte Australia – Data Analytics Job Simulation, Forage (Jul 2025)
- Applied Machine Learning in Python, Coursera (Dec 2025)
- Computer Vision, Vityarthi (Feb 2026)
- MATLAB Certified, MathWorks (Aug 2023)
- Python Essentials, Vityarthi (Dec 2023)
- Cloud Computing, NPTEL (May 2025)